Chapter 1 Foundations and Models

1.1 Solid-state Hamiltonian

In the realm of solid state physics, the many-body Schrödinger equation plays a crucial role in understanding the behavior of electrons and ions in crystalline materials. This equation provides a fundamental framework for describing the quantum mechanical interactions between multiple electrons and ions within a solid. The Schrödinger Hamiltonian for a system consisting of N_{el} electrons and N_{ion} ions — in its full beauty — reads

$$\begin{aligned}\mathcal{H} = &-\frac{\hbar^2}{2M}\sum_{i}^{N_{\text{ion}}}\nabla^2_{\mathbf{R}_i} - \frac{\hbar^2}{2m}\sum_{i}^{N_{\text{el}}}\nabla^2_{\mathbf{r}_i} + \frac{1}{2}\sum_{i\neq j}^{N_{\text{ion}}}\frac{Z_iZ_je^2}{|\mathbf{R}_i-\mathbf{R}_j|} + \\ &+\frac{1}{2}\sum_{i\neq j}^{N_{\text{el}}}\frac{e^2}{|\mathbf{r}_i-\mathbf{r}_j|} - \sum_{i}^{N_{\text{el}}}\sum_{j}^{N_{\text{ion}}}\frac{Z_je^2}{|\mathbf{r}_i-\mathbf{R}_j|} + \mathcal{V}_{\text{R}},\end{aligned} \tag{1.1}$$

where $\mathbf{r}_i$ ($\mathbf{R}_i$) are the positions of the electrons (ions), m (M) their respective masses and Z_i the atomic number of the atoms from which the crystal is constructed. The Hamiltonian above can be described in a very simple manner: the first two terms describe the kinetic energy of the ions and electrons, whereas the third and fourth terms denote the mutual Coulomb interaction between all ions (third term) and all electrons (fourth term). The second to last term corresponds to the coupling between the ions and electrons and finally $\mathcal{V}_{\text{R}}$ expresses any relativistic effects[1]. The "only" thing left to do is solve the (time-independent) Schrödinger equation using the Hamiltonian of Eq. (1.1)

$$\mathcal{H}\psi(\{\mathbf{r}_i\},\{\mathbf{R}_i\}) = E\psi(\{\mathbf{r}_i\},\{\mathbf{R}_i\}), \tag{1.2}$$

which is easier said than done. The primary problem one faces by trying to do exactly that is the sheer number of particles present in a crystalline material. To be more precise, the wavefunction needed to fulfil the eigenvalue equation above would have to depend on all the degrees of freedom of $\mathcal{O}(10^{23})$ electrons and atoms present in solids. The exponential growth in the number of configurations makes it computationally infeasible to directly calculate or even store this quantity, and already a very small number of electrons and atoms $\mathcal{O}(10)$ exceeds the capability of most modern supercomputers.

[1] e.g., spin-orbit coupling, relativistic mass correction, anomalous magnetic moments, relativistic Doppler shifts, Zitterbewegung, etc. An instructive example are the relativistic effects that take place in mercury: The binding orbitals of mercury are Lorentz-contracted due to the high speed of the electrons inside these orbitals. This results in a reduction in bonding strength, where ultimately mercury is forming a liquid at room temperatures [**Calvo2013**]. In most real solids however, relativistic effects are mostly negligible and this term will not play a significant role in the calculations we will be concerned with.

Thus, one goal of condensed matter physicists is not to solve the fundamental Hamiltonian but instead to construct approximate models that contain all the physics necessary while still being controllable and numerically feasible.
Usually, approaches to approximate Eq. (1.1) fall into two main categories: (i) using approximate methods to solve the full Hamiltonian, or (ii) applying (more) exact methods to solve an approximate, mathematically simpler model Hamiltonian. Density functional theory (DFT), for example, falls under the first category, whereas dynamical mean-field theory (DMFT) or the dynamical vertex approximation (DΓA) reside in category two. The latter build upon a model Hamiltonian which usually incorporates only a subset of all degrees of freedom, e.g. specific orbitals of atomic shells. In modern calculations, a combination of different tools is used to calculate physical properties of materials. For example, a DFT calculation provides a band structure, from which the most relevant orbitals are identified, whereafter more sophisticated methods, such as DMFT or DΓA can be applied. Within this thesis, we mainly focus on the dynamical vertex approximation, however, we will give a short introduction to DFT and DMFT in the following subsections and Sec. ??, respectively, to complete the picture.

1.2 Born-Oppenheimer approximation

Almost a century ago, it was recognized that the timescales of electronic and ionic motion differ tremendously due to the large mass disparity between them. An electron is approximately 1800 times lighter than a proton, and atomic nuclei in real solids — especially those containing transition-metal elements — are typically about five orders of magnitude heavier than electrons. This allows us, to a very good approximation, to solve the electronic degrees of freedom for static ionic coordinates. This approximation, called the Born-Oppenheimer approximation, is highly useful in condensed matter physics since it enables the study of electronic band structures, energy levels, and transport properties without the need to explicitly account for the complex interactions between electrons and lattice vibrations induced by the motion of ionic cores. However, there are certain phenomena, e.g. conventional phonon-mediated superconductivity, where the coupling of lattice vibrations with electrons can not be neglected. Luckily, there exist methods, like perturbation theory, which allow to circumvent this apparent issue. Within the regime of the Born-Oppenheimer approximation, the Hamiltonian of Eq. (1.1) reduces to

$$\mathcal{H}_{\text{B-O}} = -\frac{\hbar^2}{2m}\sum_{i}^{N_{el}} \nabla^2_{\mathbf{r}_i} + \frac{1}{2}\sum_{i\neq j}^{N_{el}} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} + \sum_{i}^{N_{el}} \mathcal{V}_{\text{ext}}(\mathbf{r}_i), \tag{1.3}$$

where $\mathcal{V}_{\text{ext}}$ describes any external potential, e.g. from the ions in the lattice. The other quantities appearing in Eq. (1.3) have already been introduced in Eq. (1.1). This approximation will serve as the basis for all upcoming models if not stated otherwise.
Despite this large simplification, the eigenvalues and the wavefunctions of the Hamiltonian in Eq. (1.3) are still impossible to compute for real-world solids. Most of the difficulties lie in the second term of Eq. (1.3), which describes the Coulomb interaction between pairs of electrons. Due to this mutual electromagnetic interaction, the evolution of every electron is influenced by all other electrons present. Hence, all electrons are correlated with each other. In general, these effects are of course not negligible, however in some materials[2], where the electrons are very mobile, screening

[2] e.g. in simple metals

effects take place which weaken the Coulomb interaction between the electrons. In this context, a very successful ansatz to make Eq. (1.3) treatable is to model the system as electrons interacting with an effective mean-field generated by all other electrons in its proximity. Such a mean field approach is density functional theory.

1.3 Density functional theory

The basic idea of density functional theory (DFT) is to work with a simple quantity, the electron density $\rho(\mathbf{r})$, instead of trying to solve the ab-initio Hamiltonian of Eq. (1.3) - or other, mathematically simpler variants - through complicated many-body wavefunctions. This is possible, at least for the ground state energy and its derivatives, thanks to the two Hohenberg-Kohn theorems [**Hohenberg1964**], which in essence state that (i) the ground state energy is a unique functional of the electron density $E[\rho(\mathbf{r})]$ (ii) which is minimized at the ground state density $\rho(\mathbf{r}) = \rho_0(\mathbf{r})$. This looks already quite promising, however, it lacks in some aspects: firstly, the two H-K theorems do not suggest any constructive/mathematical way to obtain the energy functional; secondly, this only holds for the ground state density. There is no way to extract any information from excited states; and lastly, the 3N-dimensional object $\psi(\{\mathbf{r}_i\})$ is compressed into a three-dimensional object $\rho(\mathbf{r})$. Consequently, an arbitrarily tiny variation in $\rho(\mathbf{r})$ might correspond to a huge change in $E[\rho(\mathbf{r})]$.
The big question is: "How does one obtain the energy functional $E[\rho(\mathbf{r})]$?". To answer this, we start by splitting up the energy functional into different contributions,

$$E[\rho(\mathbf{r})] = T[\rho(\mathbf{r})] + E_H[\rho(\mathbf{r})] + \int d^3r\, \rho(\mathbf{r})\mathcal{V}_{\text{ext}}(\mathbf{r}) + E_{XC}[\rho(\mathbf{r})]. \tag{1.4}$$

Here, $T[\rho(\mathbf{r})]$ denotes the kinetic energy of the electrons. $E_H[\rho(\mathbf{r})]$ is the electrostatic energy, given by the Hartree-term

$$E_H[\rho(\mathbf{r})] = \frac{1}{2}\int d^3r \int d^3r'\, \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}. \tag{1.5}$$

The potential term $\mathcal{V}_{\text{ext}}$ includes energies of electrons in external potentials (e.g. nuclei) and lastly, $E_{XC}[\rho(\mathbf{r})]$ includes everything else, i.e. the exchange and correlation energies. Dealing with the potential and the Hartree-term is easier compared to the treatment of the kinetic and exchange/correlation functional. Within Kohn-Sham DFT for example, XC functionals are often obtained by the local density approximation (LDA) or higher-order variants, such as different flavors of generalized gradient approximations (GGA). Stepping higher on Jacob's ladder[3], more sophisticated methods are hyper-GGA or several variants of (range-separated) hybrid functionals [**Perdew2001**].
This already sounds very promising, why should we not stop here and solely describe our materials with DFT? This question can be answered straightforwardly: If we had the exact energy functional, it should pose no issue. However, in realistic settings one can never acquire the exact energy functional and has to employ, e.g. Kohn-Sham DFT. Since those mean-field approaches assume that

[3] Refers to a hierarchical classification of density functional approximations (DFAs) used in Kohn-Sham density functional theory (DFT) [**Perdew2001**]. The ladder represents a sequence of increasingly accurate and complex approximations for the exchange-correlation energy, which is a key component of the total energy in DFT. Jacob's Ladder begins with the simplest and least accurate approximations at the lower rungs and progresses towards more sophisticated and accurate methods at the higher rungs. Each rung represents a level of approximation that includes additional physical effects and improves upon the previous level.

interactions are relatively weak and treat electrons as largely independent, they become less accurate when correlations are strong. Orbitals with higher angular momentum (e.g. d or f orbitals) are generally more localized in real space for a fixed principal quantum number, which leads to higher electronic densities and thus stronger local Coulomb interactions. However, the strength of these correlations depends not only on the orbital localization but also on the occupancy: correlations are particularly strong when these orbitals are partially filled, as is the case for transition-metal d shells or rare-earth f shells. In such situations, standard screening mechanisms are less effective, and mean-field theories tend to break down. Therefore, since e.g. K-S DFT is such a mean-field approach, it is not very successful in describing those systems. For example, DFT in LDA/GGA predicts a metallic behavior in partially filled 3d transition metal or 4f rare-earth elements, whereas experiments find them to be insulating. It also fails for Mott insulators because it treats electrons as too delocalized, missing strong local Coulomb correlations that drive the insulating state. Thus, for strongly correlated materials, sophisticated many-body methods, which do not approximate the full Born-Oppenheimer Hamiltonian of Eq. (1.3) directly — since many-body techniques tend to be numerically expensive — instead rely on model Hamiltonians. The most important example is the Hubbard Hamiltonian, the standard Hamiltonian of strongly correlated condensed matter systems, which is described in more detail in Sec. 1.5.

1.4 Wannier projection

We now want to define an effective (low-energy) subspace, where we can use more advanced many-body methods on. One commonly used approach is to obtain localized wavefunctions, also called Wannier functions, from extended Bloch wavefunctions used in DFT by Wannier projections [**Marzari1997**]. These localized Wannier functions are constructed for the bands near the Fermi energy to filter out irrelevant bands and obtain a simplified model for the low-energy physics. They can then be interpreted as basis orbitals localized on lattice sites, forming the foundation for effective model descriptions. In the extreme case, when only a single band is retained, this reduces to the one-band Hubbard model, see Sec. 1.6. The eigenstates of the Schrödinger equation in a periodic lattice are Bloch functions of the form

$$\psi_{n\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\mathbf{r}} u_n(\mathbf{r}), \tag{1.6}$$

where $u_n(\mathbf{r}+\mathbf{R}) = u_n(\mathbf{r})$ incorporates the periodicity of the lattice and in the simplest case, n denotes the index of a single isolated band. To obtain a local set of basis functions, one applies a Fourier transform to the Bloch functions,

$$w_{n\mathbf{R}}(\mathbf{r}) = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}\in BZ} e^{-i\mathbf{k}\mathbf{R}} \psi_{n\mathbf{k}}(\mathbf{r}), \tag{1.7}$$

where $\mathbf{R}$ is a lattice vector and $w_{n\mathbf{R}}(\mathbf{r})$ is a Wannier function centred around the lattice site $\mathbf{R}$. Wannier functions are an equivalent basis compared to the Bloch functions, however they are not eigenstates of the lattice Hamiltonian. Additionally, the choice for Eq. (1.7) is not unique, since the Bloch functions have a gauge freedom of a unitary transformation, which reads for a single isolated orbital n

$$\tilde{\psi}_{n\mathbf{k}}(\mathbf{r}) = e^{i\varphi_n(\mathbf{k})} \psi_{n\mathbf{k}}(\mathbf{r}). \tag{1.8}$$

Consequently, there is some degree of freedom to Wannier function construction, which can be applied to obtain Wannier functions that represent localized wavefunctions in a lattice — so-called *maximally localized* Wannier functions.

1.4.1 Multi-orbital Wannier functions

So far, we assumed a single isolated band with band index n. This does not capture real-world materials, where one usually has a "spaghetti" of entangled bands. Let us assume that the number of Bloch bands, which are encompassed in the Wannier projection, is L and these bands are well separated from the other bands in the material. In this case, the transformation of Eq. (1.8) can be generalized to

$$\tilde{\psi}_{m\mathbf{k}}(\mathbf{r}) = \sum_{n=1}^{L} U^{\mathbf{k}}_{mn} \psi_{n\mathbf{k}}(\mathbf{r}), \tag{1.9}$$

where $U^{\mathbf{k}}_{mn} \in M \times L$. The construction of the Wannier functions follows

$$w_{m\mathbf{R}}(\mathbf{r}) = \sum_{\mathbf{k}\in BZ} e^{-i\mathbf{k}\mathbf{R}} \sum_{n=1}^{L} U^{\mathbf{k}}_{mn} \psi_{n\mathbf{k}}(\mathbf{r}), \tag{1.10}$$

which allows for an additional mixing of the Bloch states. In this case, the unitary matrix $U^{\mathbf{k}}_{mn}$ encodes the gauge freedom of the Wannier projection. A popular unique "gauge fixing" of $U^{\mathbf{k}}_{mn}$ is through maximally localized Wannier functions.

1.4.2 Maximally localized Wannier functions

The idea here is to choose $U^{\mathbf{k}}_{mn}$ such that the sum of quadratic spreads of the Wannier functions is minimized [**Marzari1997**]. This spread of the set of L Wannier functions in real space is quantified by the localization functional Ω and reads

$$\Omega = \sum_{m} \left[\left\langle \mathbf{r}^2 \right\rangle_m - \langle \mathbf{r} \rangle_m^2 \right] = \sum_{m} \left[\langle w_{m\mathbf{0}} | \, \mathbf{r}^2 \, | w_{m\mathbf{0}} \rangle - | \langle w_{m\mathbf{0}} | \, \mathbf{r} \, | w_{m\mathbf{0}} \rangle |^2 \right]. \tag{1.11}$$

Applying a minimization procedure to the spread of Wannier functions results in a "gauge-fixed" $\tilde{U}^{\mathbf{k}}_{mn}$, choosing the unitary transformation uniquely. The resulting Wannier functions are then called maximally localized.

1.5 Multi-orbital Hubbard model

Now we have acquired enough tools to describe one of the most important models of strongly correlated electron systems. As already discussed at the end of Sec. 1.3, for elements with partially filled d or f orbitals, a mean-field description, as in DFT, is not enough to successfully describe microscopic and macroscopic phenomena due to strong electronic correlation.

A sophisticated and accurate treatment on the level of the Born-Oppenheimer approximation (Eq. (1.3)) is currently not feasible. Hence, the method of choice is the introduction of a mathematically simpler, but conceptually still very relevant model Hamiltonian that is able to capture the essential low-energy physics. As a result, the Hamiltonian - named for J. Hubbard - was developed.

As a starting point we choose the Hamiltonian of Eq. (1.3) in second quantization

$$\begin{aligned}\mathcal{H} = \sum_{\sigma} \int d^3 r\, \hat{\psi}^{\dagger}_{\sigma}(\mathbf{r}) \left[-\frac{\hbar^2}{2m} \nabla^2_{\mathbf{r}} + \mathcal{V}(\mathbf{r}) \right] \hat{\psi}_{\sigma}(\mathbf{r}) \\ + \frac{1}{2} \sum_{\sigma\sigma'} \int d^3 r \int d^3 r'\, \hat{\psi}^{\dagger}_{\sigma}(\mathbf{r}) \hat{\psi}^{\dagger}_{\sigma'}(\mathbf{r}') \frac{e^2}{|\mathbf{r} - \mathbf{r}'|} \hat{\psi}_{\sigma'}(\mathbf{r}') \hat{\psi}_{\sigma}(\mathbf{r}),\end{aligned} \tag{1.12}$$

where $\hat{\psi}^{(\dagger)}_{\sigma}(\mathbf{r})$ are field operators that annihilate (create) an electron with spin σ at position $\mathbf{r}$. We can expand these field operators in terms of maximally localized Wannier functions introduced in Sec. 1.4.2, yielding

$$\hat{\psi}^{(\dagger)}_{\sigma}(\mathbf{r}) = \sum_{\mathbf{R},m} w^{(*)}_{m\mathbf{R}}(\mathbf{r})\, \hat{c}^{(\dagger)}_{\mathbf{R}m\sigma}, \tag{1.13}$$

where $w_{m\mathbf{R}}$ is the (maximally localized) Wannier function of orbital m centered around $\mathbf{R}$, and $\hat{c}_{\mathbf{R}m\sigma}$ the corresponding annihilation operator. Note that the orbitals m are usually the localized 3d or 4f orbitals in transition metals or rare earth elements. With the help of Eq. (1.13), the Hamiltonian of Eq. (1.12) can be rewritten as

$$\begin{aligned}\mathcal{H} = -\sum_{\substack{\mathbf{R}\mathbf{R}' \\ mm',\sigma}} t_{mm'}(\mathbf{R}, \mathbf{R}') \hat{c}^{\dagger}_{\mathbf{R}m\sigma} \hat{c}_{\mathbf{R}'m'\sigma} \\ + \frac{1}{2} \sum_{\substack{\mathbf{R}_1\mathbf{R}_2\mathbf{R}_3\mathbf{R}_4 \\ ll'mm' \\ \sigma\sigma'}} U_{lmm'l'}(\mathbf{R}_1, \mathbf{R}_2, \mathbf{R}_3, \mathbf{R}_4) \hat{c}^{\dagger}_{\mathbf{R}_1 l\sigma} \hat{c}^{\dagger}_{\mathbf{R}_3 m'\sigma'} \hat{c}_{\mathbf{R}_4 l'\sigma'} \hat{c}_{\mathbf{R}_2 m\sigma},\end{aligned} \tag{1.14}$$

where $ll'mm'$ denote different Wannier orbitals, while $\mathbf{R}_i$ corresponds to a lattice site. Hence, the first term in the Hubbard Hamiltonian represents the kinetic energy and describes electrons hopping from orbital m' at site $\mathbf{R}'$ to orbital m at site $\mathbf{R}$. The hopping matrix element $t_{mm'}(\mathbf{R}, \mathbf{R}')$ (in units of energy) quantifies the amplitude for such a process and thus determines how easily an electron can move between the two orbitals. The second term in Eq. (1.14) encodes the Coulomb interaction between two electrons. Some of the terms of the multiorbital Hubbard Hamiltonian are visually depicted in Fig. 1.1, where a subset of effects of $t_{mm'}(\mathbf{R}, \mathbf{R}')$ and $U_{lmm'l'}(\mathbf{R}_1, \mathbf{R}_2, \mathbf{R}_3, \mathbf{R}_4)$ are shown for two lattice sites. Both the hopping matrix element $t_{mm'}(\mathbf{R}, \mathbf{R}')$ and Coulomb interaction

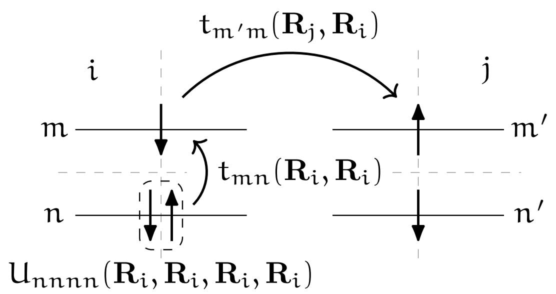

FIGURE 1.1 – Visual representation of a subset of interaction and hopping terms of Eq. (1.14). We show two lattice sites i and j with orbitals m, n and m', n', respectively.

$U_{lmm'l'}(\mathbf{R}_1, \mathbf{R}_2, \mathbf{R}_3, \mathbf{R}_4)$ are given by

$$t_{mm'}(\mathbf{R}, \mathbf{R}') = -\int d^3r\, w^*_{m\mathbf{R}}(\mathbf{r}) \left[-\frac{\hbar^2}{2m}\nabla^2_{\mathbf{r}} + \mathcal{V}(\mathbf{r}) \right] w_{m'\mathbf{R}'} \quad \text{and} \tag{1.15a}$$

$$U_{lmm'l'}(\mathbf{R}_1, \mathbf{R}_2, \mathbf{R}_3, \mathbf{R}_4) = \int d^3r \int d^3r'\, w^*_{l\mathbf{R}_1}(\mathbf{r}) w^*_{m'\mathbf{R}_3}(\mathbf{r}') \frac{e^2}{|\mathbf{r}-\mathbf{r}'|} w_{l'\mathbf{R}_4}(\mathbf{r}') w_{m\mathbf{R}_2}(\mathbf{r}). \tag{1.15b}$$

While for Eq. (1.14) we have already restricted ourselves to a subspace of orbitals, spanned by the Wannier function we picked previously, it is still too complex to be solved exactly in the general case. The complexity can be further reduced by either placing further assumptions and restrictions or by using symmetries of the Hamiltonian, which are present in the hopping and interaction terms. Note, that Eq. (1.15b) overestimates the repulsive Coulomb interaction since screening effects from electrons outside the orbital subspace are not considered. One way to partly overcome this issue is offered by the constrained Random Phase Approximation (cRPA). For details on how cRPA works, we refer the reader to, e.g. Ref. [**Zhang2022**].

1.5.1 Fourier transform of the Coulomb term

For reasons that become apparent later, it is more convenient to express the interaction term in Fourier space for diagrammatic calculations. For this we follow Ref. [**Galler2017**], where we simplify the notation by employing a compound index $i = \{\mathbf{R}_i, n_i\}$, denoting a combination of an orbital index and a real-space vector or a momentum vector, depending on the context if they are not explicitly written down. If some parameters contained in a compound index are explicitly written down, the compound index is often additionally placed to denote the remaining quantities. The real-space interaction term in Eq. (1.14) inherently depends on four lattice vectors. However, to simplify the notation using translational symmetry, one of these arguments (e.g. $\mathbf{R}_4$) can be eliminated by expressing the distances relative to the lattice site 4 rather than using their absolute positions. Therefore [**Galler2017**],

$$\hat{U} = \frac{1}{2} \sum_{\substack{\mathbf{R}_1\mathbf{R}_2\mathbf{R}_3 \\ 1234 \\ \sigma\sigma'}} U_{1234}(\mathbf{R}_1, \mathbf{R}_2, \mathbf{R}_3) \hat{c}^\dagger_{1\sigma} \hat{c}^\dagger_{3\sigma'} \hat{c}_{\mathbf{R}_4=\mathbf{0};4\sigma'} \hat{c}_{2\sigma}. \tag{1.16}$$

This quantity naturally fulfills a swapping symmetry, where the simultaneous exchange of both incoming and outgoing particles keeps it invariant, i.e.

$$U_{1234}(\mathbf{R}_1, \mathbf{R}_2, \mathbf{R}_3) = U_{2143}(\mathbf{R}_3 - \mathbf{R}_2, -\mathbf{R}_2, \mathbf{R}_1 - \mathbf{R}_2). \tag{1.17}$$

The general Fourier transform of the tensor in Eq. (1.16) with respect to the lattice positions $\mathbf{R}_i$ is defined as

$$U^{\mathbf{qkk}'}_{1234} := \sum_{\mathbf{R}_1\mathbf{R}_2\mathbf{R}_3} e^{i\mathbf{k}\mathbf{R}_1} e^{-i(\mathbf{k}-\mathbf{q})\mathbf{R}_2} e^{i(\mathbf{k}'-\mathbf{q})\mathbf{R}_3} U_{1234}(\mathbf{R}_1, \mathbf{R}_2, \mathbf{R}_3), \tag{1.18}$$

or for the full interaction operator in Eq. (1.14)

$$\hat{U} = \frac{1}{2} \sum_{\substack{\mathbf{qkk}' \\ 1234 \\ \sigma\sigma'}} U^{\mathbf{qkk}'}_{1234} \hat{c}^\dagger_{\mathbf{k}1\sigma} \hat{c}^\dagger_{\mathbf{k}'-\mathbf{q}3\sigma'} \hat{c}_{\mathbf{k}'4\sigma'} \hat{c}_{\mathbf{k}-\mathbf{q}2\sigma}, \tag{1.19}$$

where the Fourier transforms of the creation and annihilation operators are given by

$$\hat{c}^{(\dagger)}_{\mathbf{k}1\sigma} = \sum_{\mathbf{R}} e^{(-)\mathrm{i}\mathbf{k}\mathbf{R}} \hat{c}^{(\dagger)}_{\mathbf{R}1\sigma}. \tag{1.20}$$

In the case where Wannier orbitals outside of the same unit cell has negligible overlap (effectively pairing up the creation and annihilation operators at sites **0** and **R**), the interaction becomes diagonal in real space. As a result, the Coulomb matrix elements depend only on the relative positions of the sites and their Fourier transform depends at most on the momentum transfer **q**, while the explicit dependence on **k** and **k**′ vanishes,

$$U_{1234} \coloneqq U_{1234}(\mathbf{0},\mathbf{0},\mathbf{0}), \quad \text{and} \tag{1.21a}$$

$$V^{\mathbf{q}}_{1234} \coloneqq \sum_{\mathbf{R}\neq\mathbf{0}} e^{\mathrm{i}\mathbf{q}\mathbf{R}} U_{1234}(\mathbf{R},\mathbf{R},\mathbf{0}), \tag{1.21b}$$

corresponding to a possible split into a purely local interaction U_{1234} and a purely non-local interaction $V^{\mathbf{q}}_{1234}$. In this case the exchange symmetry of the incoming and outgoing electron reduces to $U_{1234} = U_{2143}$ and $V^{\mathbf{q}}_{1234} = V^{-\mathbf{q}}_{2143}$.

1.5.2 More symmetries, assumptions and the Kanamori-Hamiltonian

Despite the simplifications of the interaction tensor due to the translational symmetry, the object still has a complicated structure for a multi-orbital system. While one in principle calculate all of these elements using methods such as cRPA, there are certain symmetries and assumptions that can help reduce the amount of individual elements of $U_{1234}(\mathbf{R}_1,\mathbf{R}_2,\mathbf{R}_3,\mathbf{R}_4)$ drastically. Similar simplifications can be made for the kinetic term in Eq. (1.14). Due to the localized nature of Wannier orbitals, the Hopping amplitudes reduce (rapidly) in magnitude the further apart the orbitals are located in real space due to reduced orbital overlap. The decay of the hopping elements $t_{12}(\mathbf{R}_1,\mathbf{R}_2)$ behaves as atleast $|\mathbf{R}_1 - \mathbf{R}_2|^{-1}$[4]. Hence, for practical calculations, a very good approximation — atleast in the one-band case — can often be achieved by considering only the three closest (in distance) hopping terms: nearest (t), next-nearest (t'); and next-next-nearest (t'') neighbor, see Fig. 1.3 for an illustration. Elements of $U_{1234}(\mathbf{R}_1,\mathbf{R}_2,\mathbf{R}_3,\mathbf{R}_4)$ usually decay even faster than the hopping terms due to the overlap integral between four orbitals in Eq. (1.15b) compared to only two.

To further reduce the degrees of freedom of Eq. (1.14) and make the model more tractable, additional assumptions about the interaction can be made: (i) the Wannier functions respect SU(2) symmetry; (ii) the Wannier functions of the subset of orbitals preserve orbital symmetry; and (iii) the Coulomb interaction of electrons on different lattice sites is small compared to the on-site Coulomb interaction, e.g. $U_{1234}(\mathbf{R},\mathbf{R},\mathbf{R}) \gg U_{1234}(\mathbf{R},\mathbf{R},\mathbf{R}') \gg U_{1234}(\mathbf{R},\mathbf{R}',\mathbf{R}'')$. Let us briefly mention why these assumptions hold for systems we want to consider: (i) holds when spin-orbit coupling is ignored, the material is in the paramagnetic phase, and no external electromagnetic fields are applied [**Georges2013**]; (ii) applies to degenerate orbital subsets, like the t_{2g} and e_g orbitals in the 3d shell; and (iii) is valid as long as the Wannier functions remain localized around their

[4] More precisely, it was found that for one-dimensional and centro-symmetric three-dimensional systems the decay behaves purely exponential, i.e. $w(\mathbf{r}) \propto e^{-|\mathbf{r}|/\xi}$ [**Cloizeaux1964**]. For two- or three-dimensional insulating systems, $w(\mathbf{r}) \propto |\mathbf{r}|^{-\alpha} e^{-|\mathbf{r}|/\xi}$ [**He2001**] and for metals $w(\mathbf{r}) \propto |\mathbf{r}|^{-\alpha}$, where in this case they cannot decay exponentially [**Cornean2019**], were found.

respective lattice sites, which is generally true for d or f shells which do not (strongly) hybridize with ligand orbitals. This leads to the assumption of the interaction to be purely local, i.e. $V^{\mathbf{q}}_{1234} = 0$, since we are only keeping creation and annihilation operators in the same unit cell[5].
Under the assumptions (i)-(iii) and for a two-band model $U_{1234}(\mathbf{R}_1, \mathbf{R}_2, \mathbf{R}_3)$ is constrained to a single unit cell and has only three independent entries, commonly denoted by the intra-orbital and inter-orbital interactions U and U', respectively, and the Hund's exchange J [**Kanamori1963**, **Georges2013**]. A visual representation of the effects of U, U' and J can be found in Fig. 1.2. If the model consists of additional orbitals and/or sites, the above assumptions are not enough to reduce $U_{1234}(\mathbf{R}_1, \mathbf{R}_2, \mathbf{R}_3)$ to just three terms. These quantities are given by

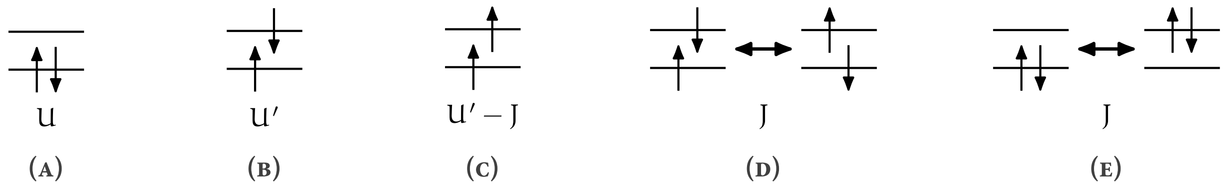

FIGURE 1.2 – Kanamori parameters U, U' and J as described in the text: (A) intra-orbital Coulomb interaction, (B)-(C) inter-orbital Coulomb interaction, (D) spin-flip of an electron pair and (E) pair hopping. Adapted from Ref. [**Galler2018**] to match the style of this thesis.

$$U = U_{1111} = \int d^3r \int d^3r' \, |w_1(\mathbf{r})|^2 \mathcal{V}_c(\mathbf{r} - \mathbf{r}') |w_1(\mathbf{r}')|^2, \tag{1.22a}$$

$$U' = U_{1122} = \int d^3r \int d^3r' \, |w_1(\mathbf{r})|^2 \mathcal{V}_c(\mathbf{r} - \mathbf{r}') |w_2(\mathbf{r}')|^2 \quad \text{and} \tag{1.22b}$$

$$J = U_{1221} = \int d^3r \int d^3r' \, w_1^*(\mathbf{r}) w_2^*(\mathbf{r}') \mathcal{V}_c(\mathbf{r} - \mathbf{r}') w_1(\mathbf{r}') w_2(\mathbf{r}), \tag{1.22c}$$

where $\mathcal{V}_c(\mathbf{r} - \mathbf{r}')$ denotes the partially screened Coulomb interaction. The terms in Eq. (1.22) correspond to the interactions of the Kanamori Hamiltonian. Since this Hamiltonian is an effective low-energy model, the high-energy electronic degrees of freedom have been integrated out, and their effect is incorporated as an effective screening of the local electron-electron repulsion. For cubic symmetry - due to constraint (ii) - there are only two independent quantities in the interaction, resulting in $U' = U - 2J$. With the help of these three parameters we can simplify the Hamiltonian of Eq. (1.14) and for the general case the so-called Kanamori-Hamiltonian [**Kanamori1963**, **Georges2013**] reads

$$\begin{aligned}
\mathcal{H}_{\text{Kanamori}} = &- \sum_{ab,\sigma} t_{ab} \hat{c}^\dagger_{a\sigma} \hat{c}_{b\sigma} + U \sum_{a} \hat{n}_{a\uparrow} \hat{n}_{a\downarrow} + \sum_{\substack{\mathbf{R},a\neq b \\ \sigma\sigma'}} (U' - J\delta_{\sigma\sigma'}) \hat{n}_{\mathbf{R}a\sigma} \hat{n}_{\mathbf{R}b\sigma'} \\
&- J \sum_{\mathbf{R},a\neq b} \hat{c}^\dagger_{\mathbf{R}a\uparrow} \hat{c}_{\mathbf{R}a\downarrow} \hat{c}^\dagger_{\mathbf{R}b\downarrow} \hat{c}_{\mathbf{R}b\uparrow} + J \sum_{\mathbf{R},a\neq b} \hat{c}^\dagger_{\mathbf{R}a\uparrow} \hat{c}^\dagger_{\mathbf{R}a\downarrow} \hat{c}_{\mathbf{R}b\downarrow} \hat{c}_{\mathbf{R}b\uparrow},
\end{aligned} \tag{1.23}$$

where $\hat{n}_{a\sigma}$ is the number operator $(\hat{c}^\dagger_{a\sigma} \hat{c}_{a\sigma})$ that counts the number of electrons in orbital a with spin σ. The Kanamori Hamiltonian is a widely used model for strongly correlated systems and it can capture a lot of low-energy physics.

[5]While it is generally true that the local interaction is stronger than $U(|\mathbf{R}_i| > 0)$, the non-local part is often not negligible. For the case of $La_3Ni_2O_7$, the non-local contributions of the interaction are believed to be long-range in-plane.

1.6 Single-band Hubbard Hamiltonian

To benchmark the code developed in the course of this thesis, we consider the well-studied case of the single-band Hubbard model, which has been extensively analyzed within the dynamical vertex approximation. We therefore briefly introduce its Hamiltonian in the following. The single-band version of Eq. (1.14) is one of the most widely used model Hamiltonians to describe strongly correlated materials and reads

$$\mathcal{H}_{\text{Hubbard}} = -\sum_{i\neq j,\sigma} t_{ij}\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma} + U\sum_{i}\hat{n}_{i\uparrow}\hat{n}_{i\downarrow} - \mu\sum_{i}(\hat{n}_{j\uparrow}+\hat{n}_{j\downarrow}), \tag{1.24}$$

where i and j denote the lattice site in the N-dimensional lattice and the hopping is spin-independent, and μ denotes the chemical potential. Since the model is reduced to a single orbital, the inter-orbital interaction terms U' and J no longer appear, as they are only defined between different orbitals.

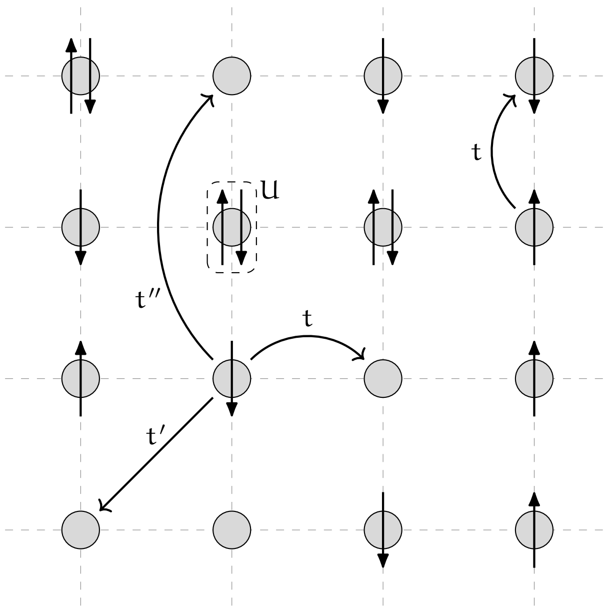

FIGURE 1.3 – Visual representation of the two-dimensional square lattice Hubbard model. Each lattice site either contains zero, one or two electrons. Every circle represents a local orbital at any site. Here, t, t' and t'' denote nearest-neighbor hopping, next-nearest-neighbor hopping and next-next-nearest-neighbor hopping, respectively. Realistic values for t, t', t'' and U are obtained from first principles or *ab-initio* methods, e.g. density functional theory (DFT) [**Pavarini2001**], ARPES [**Hashimoto2008**] or cRPA [**Werner2015**]. For e.g. cuprates, one finds that $t \approx$ 0.3-0.5 eV, $t' \approx$ -0.2t, $t'' \approx 0.1t$ and $U \approx 8t$.

This is one of the simplest forms of the general Hubbard Hamiltonian of Eq. (1.14), however it can still not be solved exactly since the kinetic term is diagonal in momentum space whereas the interaction term is diagonal in real space. A general analytical solution to the model only exists for one-dimension [**Lieb1968**] or two sites [**Jafari2008**], however recent advances allowed numerical calculations to exactly calculate the eigendecomposition of the Hubbard Hamiltonian for a small number of lattice sites [**Siro2012**]. To visually illustrate the model's description, a representation of the hopping and interaction terms of Eq. (1.24) can be seen in Fig. 1.3.

In the previous sections, we looked at the fundamental Hamiltonians and a few specific models that create the theoretical foundation for modelling strongly correlated electron systems, such as transition metal oxides (e.g. manganites, cuprates or nickelates), heavy fermion systems (e.g. compounds containing rare-earth or actinide elements), Mott insulators (e.g. V_2O_3) and many more. We have developed the fundamental ideas required to explain the distinctive behavior of particles in systems where interactions play an important role.
With the fundamentals established, in the following chapters we will introduce the mathematical framework for performing realistic material calculations within these models. The next chapter presents a comprehensive toolkit, including many-body Green's functions and diagrammatic methods such as the dynamical mean-field theory (DMFT) and the dynamical vertex approximation (DΓA), which allow us to compute physically relevant observables. The equations that follow form the core of the program developed as part of this thesis, bridging the theoretical models with their numerical implementation.